provide the requisite energy to drive machines. These engines used coal to generate steam. Coal became the principal source of energy for the industrial era. Oil became the principal energy source for transportation machines. Coal, oil and gas, all fossil fuels, fired and drove billions of machines in the world to produce goods and services which have made the global economy reach very close of $100 trillion in value today. For this, humanity mines the bosom of our planet to take out about 20 billion tonnes of material, 90 per cent of which are fossil fuels.

While machines driven by fossil fuels did a lot of good to raise living standards and make humanity grow and become rich (poverty has been banished in most of the industrial and emerging market world), the use of fossil fuels had massive side-effects. Coal produced greenhouse gases like CO_2 and polluting gases like NOx and SOx and materials like mercury. Oil produced lot of CO_2. The world produces more than 50 billion tonnes of CO_2 equivalent every year presently. The environmental issues of concern are mostly on account of the use of energy for running machines in the industrial age. Some environmental issues are caused by disturbing the natural state of the planet in carrying out agriculture operations and building infrastructure for the industrial economy.

This is the big, fat ‘environmental’ challenge for humanity.

Factories could be owned by a few and with machines working to produce goods and services, social inequality is the product of the industrial era. The increasing supply of labour (because of cheaper and universal availability of food and other industrial products) and decreasing demand of labour (because of machines increasingly taking over more and more of physical labour work) made the terms of trade weaker for labour. While the industrial economy generated considerable income and wealth, which lifted living and health standards all over the world for most people, the wealth got massively concentrated in fewer hands in the industrial economy as the income and property became more and more privately owned. The rich